AdaptiveAvgPool1d#

https://pytorch.org/docs/stable/generated/torch.nn.modules.pooling.AdaptiveA

Description:

Applies a 1D adaptive average pooling over an input signal composed of several input planes. The output size is LoutL_{out}Lout, for any input size. The number of output features is equal to the number of input planes. output_size (Union[int, tuple[int]]) – the target output size LoutL_{out}Lout. Input: (N,C,Lin)(N, C, L_{in})(N,C,Lin) or (C,Lin)(C, L_{in})(C,Lin). Output: (N,C,Lout)(N, C, L_{out})(N,C,Lout) or (C,Lout)(C, L_{out})(C,Lout), where Lout=output_sizeL_{out}=\text{output_size}Lout=output_size.

Function Signature

```
class
torch.nn.modules.pooling.AdaptiveAvgPool1d(output_size)
[source]#

Parameters

Shape:

forward(input)[source]#

Return type
```

Returns:

Tensor

Code Examples

```
>>> # target output size of 5
>>> m = nn.AdaptiveAvgPool1d(5)
>>> input = torch.randn(1, 64, 8)
>>> output = m(input)
```

Detailed Documentation

Documentation

Applies a 1D adaptive average pooling over an input signal composed of several input planes.

The output size is LoutL_{out}Lout, for any input size. The number of output features is equal to the number of input planes.

output_size (Union[int, tuple[int]]) – the target output size LoutL_{out}Lout.

Input: (N,C,Lin)(N, C, L_{in})(N,C,Lin) or (C,Lin)(C, L_{in})(C,Lin).

Output: (N,C,Lout)(N, C, L_{out})(N,C,Lout) or (C,Lout)(C, L_{out})(C,Lout), where Lout=output_sizeL_{out}=\text{output_size}Lout=output_size.

Runs the forward pass.